Quant Guild Study Plan

21 weeks · 2 hours a day · 6 days a week · starts Monday 19 October 2026

Goal. By March 2027, a public GitHub portfolio of ~16 quant finance projects plus one original capstone, each rebuilt from first principles on free data. Along the way: fluency in simulation, option pricing, volatility modelling and honest backtesting.

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Repo: github.com/giomil1998/quant-guild-study · Source lectures: Quant Guild Library by Roman Paolucci

1 · How you'll learn: the method

Watching lectures feels productive but leaves little behind. Skill comes from **producing** things: predicting, coding, getting it wrong, and explaining. Every session in this plan is built on five rules.

Rule	What it means in practice
1. Predict before you run	Before executing a cell, write one line: what do I expect to see? Being wrong is the moment learning happens.
2. Type it, never paste it	Rebuild every idea from a blank cell. If you copy code from the original notebook, close it and redo it from memory.
3. Struggle for 15 minutes, then climb the hint ladder	Docs → print/inspect variables → simplify the problem → ask Claude for a *hint* (not the answer) → only then look at a solution.
4. Explain it in writing	Every result gets a markdown cell in your own words. If you can't explain a chart, you don't understand it yet.
5. Recall beats re-reading	Start each day by answering yesterday's three questions *from memory*. Recall is what makes things stick.

How each project is structured

From project 02 on, you get a **starter notebook**: the story, the maths and a sequence of tasks with empty cells. After each task there is a **checkpoint cell** that tests your answer and prints ✓ or a hint, so you always know where you stand. When all tasks pass you add your own extension, write takeaways, and ask for a review.

Stage	You do	Time
Learn	Watch the lecture; pause and predict; take notes in your words	~1 session
Build	Work through the tasks in the starter notebook until every checkpoint passes	~3 sessions
Extend	One idea of your own, ideally on real data	~1 session
Ship	Takeaways, review with Claude, update the README and ROADMAP, push	~1 session

2 · The daily 2-hour block

Same structure every day, so no time goes on deciding what to do. Use a timer.

Time	Block	What you do
0:00 – 0:10	**Recall**	Answer yesterday's 3 questions from memory in LEARNING_LOG.md. Check yourself afterwards.
0:10 – 0:40	**Learn**	Lecture segment or reading. Pause before each result and predict it. Notes go into the notebook's markdown.
0:40 – 1:40	**Build**	Next task(s) in the starter notebook. Predict → type → run → fix. Phone in another room.
1:40 – 1:55	**Explain**	Write the interpretation cell for what you built. One-sentence takeaway.
1:55 – 2:00	**Ship**	Commit and push. Log entry: what I did, what confused me, 3 questions for tomorrow.

The weekly rhythm

Day	Focus
Mon	Learn the week's lecture(s) + first build task
Tue – Thu	Build: work through the tasks and checkpoints
Fri	Extension on real data + takeaways; ask Claude for a review
Sat	Portfolio day: fix review comments, update README/ROADMAP, push. Then watch the week's *study* lectures (no build needed)
Sun	Rest. Optional: 20-minute quiz with Claude on the week's material

Fell behind? Don't skip ahead and don't try to double up. Shorten the extension, ship what you have, and use the next buffer week. A shipped 80% project teaches more than an abandoned perfect one.

3 · Tools: PyCharm, Jupyter and git

Recommendation: use PyCharm for everything, since you already know it. Its notebook support gives you the inline charts, plus autocomplete, a variable viewer and a real debugger. If your PyCharm asks for a paid plan to run notebooks, use JupyterLab in the browser for the notebooks and keep PyCharm for the `.py` files and git.

One-time setup in PyCharm

- **File** → **Open** → `~/projects/quant-guild-study`
- **Settings** → **Project** → **Python Interpreter** → **Add Interpreter** → **Add Local** → **Select existing** → `.venv/bin/python`
- Open `notebooks/02_.../*.ipynb`. Run a cell with **Shift+Enter**. Choose the **.venv** interpreter if PyCharm asks for a kernel.
- Turn on the **Jupyter Variables** tool window: watching arrays change is the fastest way to debug.

JupyterLab fallback (Terminal)

```
cd ~/projects/quant-guild-study
source .venv/bin/activate
jupyter lab
```

The git loop: commit daily, it's your progress diary

In PyCharm: **Cmd+K** to commit (tick the files, write a message) and **Cmd+Shift+K** to push. In the terminal:

```
git status             # what changed?
git add -A
git commit -m "P02: task 3, edge decomposition"
git push
```

A steady run of daily commits on your GitHub profile is itself a signal recruiters notice.

Where code lives

Location	Put here
`notebooks/NN_topic/`	The project: story, maths, code, charts, takeaways
`src/qgs/`	Functions you reuse in 2+ projects (data loading, metrics, simulators). Edit these as .py files in PyCharm
`study/`	This plan and LEARNING_LOG.md

4 · Using Claude as your teacher

Open a Claude Code session in the project folder. These prompts cover almost everything you'll need:

When	Say
Start of a session	*"Start my session: week 1, day 2."* → you get today's tasks and the 3 recall questions
Stuck	*"I'm stuck on P02 task 3. Here's my code and the error. Give me a hint, not the answer."*
Concept unclear	*"Explain the law of total expectation simply, then give me a 10-minute exercise."*
Finished a project	*"Review my P02 notebook like a quant interviewer. Be strict."*
New starter notebook	*"I've finished P02. Set up P03 for me."*
Sunday quiz	*"Quiz me on this week's material: 5 questions, one at a time."*
Off-track	*"I've missed 3 days. Re-plan the rest of this phase."*

The deal: I write the starter notebooks, checkpoints, hints and reviews. **You write every line of project code.** If you ask for a full solution, I'll first ask what you've tried. Bypassing the struggle only cheats your future self.

5 · The 21-week plan

__Build__ = full portfolio project with a starter notebook. __Study__ = watch, plus an optional 20-minute scratch exercise; no portfolio entry. __Research__ = a sprint on your own data, kept in a private repo. Numbers are Quant Guild lecture numbers (see ROADMAP.md for links).

Phase 1 · Edge, randomness and ruin

What an edge is, how to measure it, and why bet size decides whether you survive it.

Wk	Dates	Build project	Lectures	Focus	Study
1	19 Oct – 24 Oct	**P02 · Statistics problems in trading**	3	Estimation error, non-stationarity, decomposing an edge. *Extension:* SPY's edge with a bootstrap confidence interval.	4
2	26 Oct – 31 Oct	**R1 · Research sprint (private)**	—	One-week feasibility study on your own FX options data, in a separate private repo: data inventory, matching, a first comparison. Then decide whether to go deeper.	—
3	02 Nov – 07 Nov	**P03 · Gambler's ruin**	28, 26	Ruin probability: formula vs simulation. *Extension:* ruin risk of a leveraged SPY / EUR/USD strategy.	34
4	09 Nov – 14 Nov	**P04 · Kelly, ergodicity, volatility drag**	36, 81, 136	Growth-optimal sizing, time vs ensemble averages. *Extension:* Kelly leverage on SPY, full vs half Kelly.	46, 60

Milestone M1 · 4 projects live, profile README published, repo pinned

Phase 2 · Probability and simulation toolkit

The Monte Carlo and stochastic-process tools every later module uses.

Wk	Dates	Build project	Lectures	Focus	Study
5	16 Nov – 21 Nov	**P05 · Monte Carlo foundations**	1, 33, 61	Inverse transform, law of large numbers, CLT, error bars on simulations.	2
6	23 Nov – 28 Nov	**P06 · Brownian motion and GBM vs reality**	59, 135, 29	Simulate ABM/GBM, Itô's lemma. *Extension:* calibrate GBM to SPY & EUR/USD and test it against fat tails (QQ plots).	31, 37
7	30 Nov – 05 Dec	**P07 · Jumps and clustering**	82, 94	Poisson and Hawkes processes. *Extension:* fit them to the arrival times of large-move days.	71, 49

Phase 3 · Measuring performance and backtesting honestly

How to tell skill from luck, and why most backtests lie.

Wk	Dates	Build project	Lectures	Focus	Study
8	07 Dec – 12 Dec	**P08 · Sharpe, CAGR, luck vs skill**	101, 125, 48, 96	Sharpe ratio confidence intervals, deflated Sharpe, minimum track-record length.	112
9	14 Dec – 19 Dec	**P09 · Backtesting honestly**	97, 77, 93	A small walk-forward backtester with costs; a demonstration of overfitting.	58, 118
10	21 Dec – 26 Dec	**Review and consolidation**	—	Move reusable code into `qgs`, rewrite the README, mid-point portfolio review with Claude.	—
11	28 Dec – 02 Jan	**Holiday buffer**	—	Catch up on unfinished tasks, or rest. Both count.	—

Milestone M2 · 8 projects live, helpers refactored, README case-study style

Phase 4 · Option pricing

Black-Scholes from several angles, then beyond it.

Wk	Dates	Build project	Lectures	Focus	Study
12	04 Jan – 09 Jan	**P10 · Black-Scholes three ways**	83, 19, 73	Closed form, Monte Carlo, risk-neutral argument; convergence and error.	18, 6, 67
13	11 Jan – 16 Jan	**P11 · Finite differences and exotics**	38, 32	Explicit / implicit / Crank-Nicolson; barrier and Asian options.	—
14	18 Jan – 23 Jan	**P12 · Heston and FFT pricing**	39, 89	Carr-Madan FFT; the smile Heston generates; implied-vol inversion.	107

Phase 5 · Trading options and volatility (your home ground)

Where your FX options research gives you an edge over most portfolios.

Wk	Dates	Build project	Lectures	Focus	Study
15	25 Jan – 30 Jan	**P13 · Delta hedging and Greeks P&L**	9, 11	Hedging error vs rebalancing frequency; gamma/theta P&L attribution.	7
16	01 Feb – 06 Feb	**P14 · Implied-vol surface from free data**	30, 23	Build SPY's IV surface from yfinance chains: smile, skew, term structure.	24
17	08 Feb – 13 Feb	**P15 · Volatility risk premium**	119, 43	VIX-implied vs realised SPY variance since 1990; implied-vol crush around earnings.	106, 113

Milestone M3 · 15 projects; options section is the portfolio's centrepiece

Phase 6 · Time series and portfolios

Volatility models, regimes and portfolio construction.

Wk	Dates	Build project	Lectures	Focus	Study
18	15 Feb – 20 Feb	**P16 · GARCH and regimes**	44, 47, 51	GARCH on EUR/USD; hidden Markov regimes on SPY.	92
19	22 Feb – 27 Feb	**P17 · Portfolio construction**	20, 100, 17	Why mean-variance is unstable; Black-Litterman; PCA on ETF returns.	78, 128

Capstone

One original project that ties the pieces together and shows how you think.

Wk	Dates	Build project	Lectures	Focus	Study
20	01 Mar – 06 Mar	**Capstone: design and build**	your choice	Ideas: FX vs equity volatility risk premium · Kalman-filter mean reversion (95) · tail-risk hedging (126, 138).	—
21	08 Mar – 13 Mar	**Capstone: finish and write up**	—	Polished notebook, README case study, short LinkedIn/blog post.	—

Milestone M4 · Capstone published; portfolio ready to show recruiters

After week 21

Module 8 (machine learning and market making: lectures 13, 16, 35, 63, 65, 69) and the remaining lectures in ROADMAP.md, at your own pace. By then you'll know the workflow well enough to plan it yourself.

6 · Definition of done

A project counts as shipped only when every box is ticked. This is what makes it portfolio-grade rather than homework.

✓	Check
☐	All checkpoint cells pass; the notebook runs top-to-bottom on a fresh kernel (*Restart & Run All*)
☐	Opening cell: the question, a credit link to the lecture, a 'what's new vs the original' table
☐	Every chart has a title, axis labels, and a markdown cell interpreting it
☐	At least one extension on real free data, or a rigorous test the original lacks
☐	Takeaways section in your own words (3–5 points)
☐	Review by Claude done and comments addressed
☐	README project table + ROADMAP status updated; committed and pushed with outputs saved

Milestone tracker

	Milestone	Target	Done on
☐	**M1** · 4 projects live, profile README, repo pinned	14 Nov	
☐	**M2** · 8 projects, helpers refactored, README rewritten	26 Dec	
☐	**M3** · 15 projects, options section complete	13 Feb	
☐	**M4** · Capstone published, portfolio ready for recruiters	13 Mar	

Skills you'll be able to show at the end

Area	Skills
Python	Vectorised NumPy/pandas, clean plotting, reusable modules, git workflow, reproducible notebooks
Probability	Monte Carlo with error bars, stochastic processes (BM, GBM, Poisson, Hawkes), Itô calculus intuition
Options	BS / MC / PDE / FFT pricing, Greeks and hedging P&L, implied-vol surfaces, volatility risk premium
Statistics	Estimation error, non-stationarity, bootstrap, Sharpe inference, GARCH, HMMs
Research	Honest backtesting, luck-vs-skill tests, written communication of results